

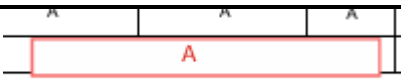
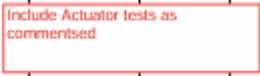
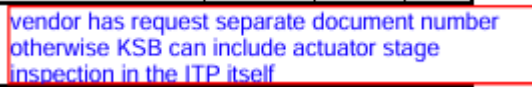
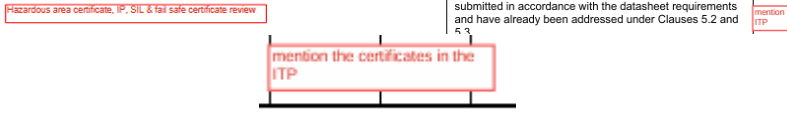
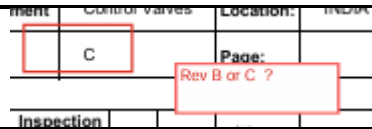
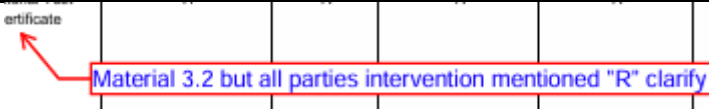
 	ADNOC ONSHORE PROJECT NO. P30339B EPC for SE AiP5 Project (On plot) - ASAB/SAHIL (Package 1) EPC AGREEMENT NO: 4700025071	
COMMENT RESOLUTION SHEET (CRS)		

ARCHIRODON TRANSMITTAL NUMBER	ADNOC ONSHORE TRANSMITTAL NUMBER	DATE	REVIEW CODE	DOCUMENT NO.	DOCUMENT TITLE	REVISION
CTA-VD-4700025071-007471	ATC-VD-4700025071-002985	26-AUG-2026	2	P30339B-30-99-52-4607	INSPECTION AND TEST PLAN FOR CONTROL VALVES	1
ADNOC ONSHORE/ARCO - REVIEW BY				VENDOR / SUBCON - RESPONSE BY		
NAME	Anish Raj			NAME	Stinu Stephen	
POSITION	Quality Engineer			DISCIPLINE	Project Engineer	
DATE	26-AUG-2026			DATE	02-SEP-2026	

SNO.	PAGE / SECTION	AON COMMENTS	ARCO COMMENTS	VENDOR / SUBCON RESPONSE	ARCO COMMENT ON VENDOR / SUBCON RESPONSE	AON COMMENTS ON ARCO RESPONSE	STATUS (OPEN / CLOSED)
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REVISION 1

1			Piston Seal is verified during Actuator Leak test or actuator pressure test ?	Kindly note that, piston seal is verified during actuator leak test.	Noted		
2			Noted. Thrust verification for actuator shall be done as per thrust calculation	Please note that we are internally verifying the actuator thrust based on the actuator thrust calculation sheets submitted under document numbers P30339B-14-01-52-4706, P30339B-14-01-52-4710, and P30339B-16-01-52-4639. These documents have already been Code 1 approved. Kindly note that separate thrust verification shall not be applicable for the control valve actuators.			
3			ISO 19840 / Visual and ISO 4628-1 to 6	Incorporated			Closed

4				Incorporated	Noted	Closed
REVISION B RESUB						
1				Refer the response on rows 7 & 8		
2				Actuator test details are already incorporated in the same ITP itself.		
3				Please note that the Hazardous Area Certificate and Degree of Protection (IP) Certificate have been included under Clause 5.3. The SIL Certificate is not applicable for the offered scope; hence, the same cannot be provided. We have already included Fail safe action check in Clause 7.8 and same will be reported in the valve TC		
4				Since Revision B has been approved by ADNOC with Code 2, the ITP has been resubmitted under Revision 1.		
5				Kindly note that Heat treatment shall be a review point only. Please note that the foundries shall be operational during 24 hours and heat treatment activities will be performed based on the furnace load. Hence it is not practically feasible to arrange inspection call & offer Heat treatment witness to TPI & will delay the process. Also same is a time-consuming process & arranging witness for the same is not practically feasible. We shall provide heat treatment details in Material test certificate & same shall be submitted for review to TPI during inspection. This is the procedure we are following for all 3.2 orders including previous ADNOC projects. Also, this point was not agreed during the TBE stage & hence we cannot accept the same. Please waive off this comment. Please note that witness by all parties for the chemical and mechanical testing has already been included under Clause 3.3 & 3.4 to comply with the 3.2 certification requirement.		
6				Please note that, during the raw material identification stage, the heat number punched on the rod will be verified and correlated by the TPI with the corresponding Rod Material Test Certificate (MTC) to ensure complete material traceability & same is update in Clause 3.1.		
7				- Kindly note that the actuator shell pressure test has been incorporated under Clause 7.3. The actuator leak test witnessing requirement has also been incorporated under the same clause - Please note that there is no separate piston seal leak test applicable. Any leakage through the piston seal would adversely affect the actuator performance and would be identified during the		

									actuator leak test & valve performance checking as per clause 7.2. Hence, a separate piston seal leak test is not required.			
									3. Since torque test is not applicable for this project, same is not included.			
									4. The requirement has been incorporated under Clause 7.3.			
8									1. Kindly note that the Spring Material Test Certificate will be provided by the spring supplier in accordance with the manufacturer's standard for your review. Same is added in Clause 5.2.			
									2. Please note that, during the valve calibration process, multiple opening and closing cycles are already performed to verify the proper operation and functionality of the valve assembly. Witnessing 10 opening and closing cycles for 100% of the valves during final inspection will considerably increase the inspection duration without providing additional technical benefit. Furthermore, this requirement is not specified in the applicable AGES documents or the project specification. Hence, we kindly request you to waive this comment.			
9									Kindly note that the positioner calibration is carried out as part of the valve performance test, and the same is indicated in the remarks section of Clause 7.7. Therefore, there is no separate positioner calibration test applicable.			
10									Noted & updated as per the discussion with Archo team.			

REVISION B

1			  ADNOC ONSHORE PROJECT NO. P30339B EPC for SEP5 Project (On plot) - ASAB/SAHIL (Package 1) EPC AGREEMENT NO: 4700025071 COMMENT RESOLUTION SHEET (CRS) ARCHIRODON TRANSMITTAL NUMBER P30339B-ARCO-KSB-000008 ADNOC ONSHORE TRANSMITTAL NUMBER DATE 28-11-2024 REVIEW CODE 2 DOCUMENT NO. P30339B-30-99-52-4607 DOCUMENT TITLE INSPECTION AND TEST PLAN FOR CONTROL VALVES Follow the correct transmittal number for both ARCO and ADNOC Follow the ADNOC transmittal title ASAB/SAHIL	Noted & updated			
2			VALVES ASAB/SAHIL	Incorporated			
3			ASAB/SAHIL Correct the same in the all pages TEST PLAN FOR CONTROL VALVES	Incorporated			

4			include P30339B-30-99-52-4608 inspection and test plan for actuator and pressure regulator P30339B-30-99-95-461 CONTROL VALVES FACTORY ACCEPTANCE TEST PROCEDURE (FAT PROCEDURE) (VALVE + ACTUATOR)-ASAB/SAHIL	Incorporated																	
5			Impact test shall carried out for CS forging and casting at -29 deg. C <previous comment not incorpora	As per the discussion with the customer, the comment is waived off by the ARCO team With reference to the discussion with our QC team please note that for A 216 WCC material impact test in not required , kindly proceed accordingly. Thanks & Regards, Aravind B Pillai Procurement Department Archirodon Construction (Overseas) Co. Ltd.																	
6			ASTM G48-A (type 22Cr) - 25 degree Celsius duration- 24 hours. Acceptance criteria: No pitting at 20X magnification and weight loss ASTM G48-A (type 25Cr) - 50 degree Celsius duration- 24 hours. Acceptance criteria: No pitting at 20X magnification and weight loss ASTM 28-A - 120 degree Celsius duration - 120 hours. Acceptance criteria : No pitting at 50X magnification and weight loss 0.457 mm	As per AGES-SP-09-003, Clause 11.8.5, the corrosion test acceptance criterion for valve base materials is specified as less than 0.914 mm/year. There is no requirement in AGES-SP-09-003 to meet a more stringent limit of 0.457 mm/year for valve base materials. Applying the corrosion limit specified in AGES-SP-09-015 (0.457 mm/year) for base material is not technically applicable, as this document governs clad piping components and not valves. Furthermore, imposing this requirement would be beyond the contractual scope and specification requirements for valves, and is not supported by the project governing valve specification. Accordingly, we confirm that the corrosion test results will be assessed against the < 0.914 mm/year acceptance criterion, which is fully compliant with the project specification and contractual requirements. Hence kindly confirm your acceptance.																	
7			Steel - MPT for Carbon steel Alloy Steel (Wet Fluorescent) DPT for alloy steel	Kindly note that, for alloy steel RM, MPT will be done and DPT is not practical.																	
8			As commented previously Vendor shall add as per AGES-Sp-7-004, Table 14.1 inspection and tests requirement: Steel and welding imperfections / Welding flux removal Inspection with reference to: ISO 8501-3 / using Distilled water and litmus paper	As per the discussion with ARCO team, this comment is waived off																	
9			<table><tr><td>8.13</td><td>Surface preparation and Painting</td><td>Final DFT / Visual Examination</td><td>100% of surfaces</td><td>Visual and 8501-3 ISO 4628-1 to 6</td><td>chipping, contamination, orange peel, cracking, blistering, rust, damages and any other defects(Witness applicable for only final DFT,other Stages(primer,intermediate dft shall be reviewed only)</td><td>RFI /</td></tr><tr><td>8.14</td><td>Surface preparation and Painting</td><td>(if applicable) Curing of Inorganic Zinc Silicate Primer using MEK solvent Rub Test</td><td>Spot checks</td><td>ASTM D4752</td><td>Conform to Level 5</td><td>RFI /</td></tr></table> Vendor shall reference all applicable standards in accordance with AGES-SP-7-004. ISO 19840 shall be explicitly referenced as stated in Table 14.1 (Inspection and Tests Requirement). Any other internal vendor standards shall either be attached for information purposes only or removed from the submission	8.13	Surface preparation and Painting	Final DFT / Visual Examination	100% of surfaces	Visual and 8501-3 ISO 4628-1 to 6	chipping, contamination, orange peel, cracking, blistering, rust, damages and any other defects(Witness applicable for only final DFT,other Stages(primer,intermediate dft shall be reviewed only)	RFI /	8.14	Surface preparation and Painting	(if applicable) Curing of Inorganic Zinc Silicate Primer using MEK solvent Rub Test	Spot checks	ASTM D4752	Conform to Level 5	RFI /	Please note that, if ISO 19840 to be followed, it will have practical issues while measuring DFT values and may lead to paint reworks. Hence kindly ignore this comment.			
8.13	Surface preparation and Painting	Final DFT / Visual Examination	100% of surfaces	Visual and 8501-3 ISO 4628-1 to 6	chipping, contamination, orange peel, cracking, blistering, rust, damages and any other defects(Witness applicable for only final DFT,other Stages(primer,intermediate dft shall be reviewed only)	RFI /															
8.14	Surface preparation and Painting	(if applicable) Curing of Inorganic Zinc Silicate Primer using MEK solvent Rub Test	Spot checks	ASTM D4752	Conform to Level 5	RFI /															
10			H A A A	Incorporated																	

REVISION A RESUB

1	3		Reference standards shall be separate table	Incorporated			
2	3			The GAD has been submitted for approval.			

			GA drawing not submitted & approved. ITP shall be resubmitted only after GA drawing approval																									
3	3		ADNOC approved MS word template to be used for this sheet. test procedure. Shall be modified to table format.	Incorporated																								
4	3		Add Actuator ITP & test procedure.	Please note that there is no separate ITP for the actuator.																								
5	3		Attach area wise (Asab & Sahil) valve list with Size, MOC, NDE requirements ext.	The same has been incorporated. Kindly refer to the attached Annexure-1 for details.																								
6	4		For all 3.2 material source inspection including "Sample taking (heat transfer), Chemical, Mechanical, NDE, Corrosion, visual & dimension shall be witnessed by ARCO & ADNOC along with 3.2 TPIA (as per AGES-SP-13-002, table 3 inspection class I & II). "Kindly specify the Casting/Forging manufacture location or inspection location in the remarks" - <applicable for all the 3.2 material inspection activities>	Please note that chemical, mechanical, NDE, corrosion, visual, and dimensional witnessing have not been added, as dimensional witnessing is not mandatory as per Clause 3.2 and was also not required in the previous Swing projects.																								
7	4		<table><tr><td>Sub Vendor</td><td>KSB</td><td>TPIA (applicable where 3.2 certification required)</td><td>Archirodon</td><td>ADNOC Onshore</td></tr><tr><td>-</td><td>H</td><td>-</td><td>A</td><td>A</td></tr><tr><td>-</td><td>H</td><td>-</td><td>A</td><td>A</td></tr><tr><td>-</td><td>H</td><td>H</td><td>H</td><td>H</td></tr></table>	Sub Vendor	KSB	TPIA (applicable where 3.2 certification required)	Archirodon	ADNOC Onshore	-	H	-	A	A	-	H	-	A	A	-	H	H	H	H	Incorporated				
Sub Vendor	KSB	TPIA (applicable where 3.2 certification required)	Archirodon	ADNOC Onshore																								
-	H	-	A	A																								
-	H	-	A	A																								
-	H	H	H	H																								
8	4		Approved data sheet / A 995 Gr.8A (CD3M/MNCu/N) / A 494 CU5MCu/C(N08828 / AGES-SP-04-002, AGES-SP-09-003, AGES-SP-07-003/ NACE MR0175 / Approved GAD For DSS material G48 method A shall be acceptance criteria. For Inconel material G28 method A shall be acceptance criteria. include acceptance criteria Material Test certificate H	Incorporated for Inconel material																								
9	4		<table><tr><td>Material Test certificate</td><td>H</td><td>F</td></tr><tr><td>Material Test Certificate</td><td>H</td><td>V</td></tr><tr><td>Material Test Certificate</td><td>H</td><td>V</td></tr><tr><td>Material Test Certificate</td><td>H</td><td>V</td></tr><tr><td>include acceptance criteria</td><td>H</td><td>V</td></tr><tr><td>Material Test certificate</td><td>H</td><td>V</td></tr><tr><td>Material Test certificate</td><td>H</td><td>V</td></tr></table> BS EN 10204 - 3.2	Material Test certificate	H	F	Material Test Certificate	H	V	Material Test Certificate	H	V	Material Test Certificate	H	V	include acceptance criteria	H	V	Material Test certificate	H	V	Material Test certificate	H	V	Incorporated			
Material Test certificate	H	F																										
Material Test Certificate	H	V																										
Material Test Certificate	H	V																										
Material Test Certificate	H	V																										
include acceptance criteria	H	V																										
Material Test certificate	H	V																										
Material Test certificate	H	V																										

10	4			Please note that the impact test shall be carried out at -46 °C in accordance with the applicable ASTM standard and in line with the previous Swing Project.			
11	5			The NDE requirements are clearly defined and separately specified for stainless steel and carbon steel items.			
12	5			Incorporated			
13	5			Added			
14	6			The same is applicable for cladding, in line with BB08E.			
15	6			Incorporated			
16	6			The actuator details shall be duly certified in the Valve TC.			
17	6			Please note that this requirement has already been addressed and included in Clause 7.2.			
18	7			As the actuator is of our own make, the relevant details shall be duly certified in the Valve TC.			
19	7			Kindly provide the document number			
20	7			Incorporated			

21	7		AGES-SP-07-004 Requirements/PQT shall be done on sample pad of dimension 300mm*300mm RFI / Rep Vendor to add: ASTM D4228	Incorporated															
22	8		Page 1 of 6 KSB M KSB M 04/08 Vendor to add: Steel and welding imperfections / Welding flux removal Inspection with reference to: ISO 8501-3 / using Distilled water and litmus paper	The same is not applicable, and we are facing difficulties in incorporating it, as this is not a standard practice followed by us.															
23	9		ISO 19840 / Visual and ISO 4628-1 to 6 mination 100% of surfaces Visual and SSPC-PA2	Please note that ISO 4628 (Visual) has already been included. ISO 19840 has not been added, as we are following SSPC-PA 2 instead of ISO 19840.															
24	9		W (10%) W (10%) W (10%) Visual Inspection 100 %	Incorporated															
25	9		<table><tr><td>W (10%)</td><td>W (10%)</td><td>W (10%)</td></tr><tr><td>W (10%)</td><td>W (10%)</td><td>W (10%)</td></tr><tr><td>H</td><td>H</td><td>R</td></tr><tr><td colspan="3">A</td></tr></table>	W (10%)	W (10%)	W (10%)	W (10%)	W (10%)	W (10%)	H	H	R	A			Incorporated			
W (10%)	W (10%)	W (10%)																	
W (10%)	W (10%)	W (10%)																	
H	H	R																	
A																			
26	9		g. Regularly completed as per the requirements above, and according to the Inspection Release Note issued. The Certification requirements (EN 10204 - 3.2) Shall be applicable to the complete assembled control valves not only material. So the vendor shall include separate inspection stage for the 3.2 certification for equipment from 3.2 TPI. (TYP)	Added separate clause															
27	9		Hazardous area certificate, IP, SIL & fail safe certificate review	Please note that the applicable certificates shall be submitted in accordance with the datasheet requirements and have already been addressed under Clauses 5.2 and 5.3.															

REVISION A

1	3		Criticality rating 1, inspection class 1	Incorporated			
2	3		Update the PO number	Incorporated			
3	3		Document number shall be 30-99-52-4607	Incorporated			
4	3		update the equipment details	Incorporated			
5	3		Update the Location	Incorporated			
6	3		Update the Page number	Incorporated			

7	3		Notification Period: Minimum Include Extent of Examination in between Activity description and Reference document.	Incorporated			
8	3		ACT.NO.: list out all procedures as per approved VDRL	Please refer the attached reference documents.			
9	3		Procedure Code H	Incorporated			
10	5		Version 2024 AGES-SP-07-004 so generic be specific when cladding feasible why not ??	Please clarify the comment.			
11	5		list out the accessories	Incorporated			
12	5		where ever low, high carbon stainless steel used, Carbon detection is	Please note that, in line with AGES requirements, carbon detection is not mandatory for this scope.			
13	5		W W (10%) W (10%)	In line with the agreement with the customer, a random witness (W) at 10% has been considered and reflected in the ITP accordingly.			
14	6		body shell test to be included with witness for archirodon and ADNOC	Kindly refer clause 6.1			
15	6		ipition Refe (Valve +Actuator)	Specified in clause 7 (header)			
16	6		other than shell test other applicable leak test shall be with actuator Verify occur	Clause 7 header indicates valve with actuator.			
17	6		leakage rate shall be addressed inline with approved data sheet	Noted			
18	6		As per AGES-SP-07-004, the governing specification referenced in the MR and ITP, the vendor shall incorporate the minimum requirements specified in Section 7.13 to 7.17 into the ITP for Painting Activities, in line with the attached sample ITP (refer to page 8), ensuring inclusion of all relevant standards and referen	Please note that we have prepared the ITP in line with previously executed ADNOC projects, namely EPCM for Project SWING (ADNOC Onshore Project No.: P30332) and EPCM for	NOT Accepted. The vendor shall comply with		

				BAB & BU HASA (ADNOC Onshore Project No.: P30350). Kindly confirm your acceptance. KSB MIL Reply dt 04/11/25 – Noted & painting related activities inline with received sample ITP as applicable for the offered valves have been incorporated in the revised ITP.	the requirements of AGES-SP-7-004.								
19	7		release shall be after MRB	MRB submission will be after valve despatch.									
20	7		NACE and Hazardous area certification review as applicable	Noted									
21	7		Provide a list of reference document GA drawing, data sheet, test procedures, painting procedure etc.	Kindly refer the attached reference document.									
22	1		ADNOC ONSHORE DOCUMENT NO. P30339B 30-99-52-4607	Noted & updated									
23	2		Project No.: P30339B ADNOC Onshore Doc. No.: P30339B 30-99-52-4607 	Noted & updated									
24	4		Rev A Mention where the inspection is happening in this case mention (India) ASAB/SAHIL (PACKAGE 1) <table><tr><td>Control Valves</td><td>Location:</td><td>ASAB/SAHIL</td></tr><tr><td>1</td><td>Page:</td><td>Page 1 of 5</td></tr></table> 	Control Valves	Location:	ASAB/SAHIL	1	Page:	Page 1 of 5	Noted & updated			
Control Valves	Location:	ASAB/SAHIL											
1	Page:	Page 1 of 5											
25	7		Procedure / AGES-SP-07-004 Procedure / AGES-SP-07-004 The vendor is requested to provide detailed explanations for the painting activities as indicated on Page 9. As noted in CRS ARCO's response, the submission is not accepted; the vendor shall confirm that all requirements are fully incorporated within the ITP as per AGES-SP-7-004, and ensuring inclusion of all relevant standards and clear reference to the Company-approved painting procedure document number.	Noted & ITP has been revised incorporating the painting requirements in line with AGES -SP-7-004 as per Clause 8 as applicable for the offered valves. Please refer the following points (SI nos 26 - 33) for our clarifications on the painting ITP.									
26	ITP Painting Activities "AGES-SP-07-004" - Point No 1		ISO 12944-9, category CX - Fulfilment of CX testing	Please note that we are exactly following ADNOC scheme as per AGES spec for the applicable materials, insulation & temperature range (For actuator, we have considered scheme for carbon steel material and uninsulated category) and as per our understanding, if ADNOC AGES scheme is exactly followed, there is no need for separate CX compliance. Also, it is an onshore project and hence CX seems not relevant.									
27	ITP Painting Activities		Pre-production / Repair Qualification Test	Kindly note that PQT shall be done as per AGES-SP-07-004 on sample pad of dimension 300 mm X 300 mm for each paint system and									

	"AGES-SP-07-004" - Point No 2			adhesion test shall be done for qualifying the same. We shall perform the same & report shall be submitted for review. During painting inspection, adhesion test on sample pad shall be offered for witness.			
28	ITP Painting Activities "AGES-SP-07-004" - Point No 3		Paint Material inspection (Verify storage, container integrity, and shelf life) - Storage +15° C to +25° C Conductivity of Abrasives - $\leq 150 \mu\text{mS/cm}$ at 20° C	Kindly note that the storage of paint material shall be done as per the recommendation by paint manufacturer in line with paint datasheet. Also conductivity of abrasives shall be tested at ambient temperature			
29	ITP Painting Activities "AGES-SP-07-004" - Point No 4		Steel and welding imperfections / Welding flux removal Inspection	Same is not applicable for the offered valves & hence not included in the ITP.			
30	ITP Painting Activities "AGES-SP-07-004" - Point No 8		Protection of fittings and accessories	Kindly note that during surface preparation, fittings & accessories will not be mounted & hence this clause is not relevant for the offered valves and hence not included in ITP.			
31	ITP Painting Activities "AGES-SP-07-004" - Point No 10		Power tool cleaning	Please note that power tool cleaning is not applicable for the offered valves & hence this clause is not relevant for the offered valves and hence not included in ITP			
32	ITP Painting Activities "AGES-SP-07-004" - Point No 14		Apply a Stripe Coat by Brush	Please note that stripe coat by brush is not applicable for the offered valves & hence this clause is not relevant for the offered valves and hence not included in ITP			
33	ITP Painting Activities "AGES-SP-07-004" - Point No 16		DFT Measurement – reference standard ISO 19840	Please note that reference standard for DFT measurement shall be SSPC PA2			

34	3.2 Witness – scope addition for TPIA for FAT		by Intervention TPIA (applicable where 2.2 certification required) – – – –	Noted & updated			
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Review Status Code	Review Response	SUBCONTRACTOR/VENDOR CHECKED BY	DATE	SIGNATURE
1	Approved / No Comments	Ginoy Shaju	02-SEP-2026	
2	Approved with comments – Resubmit			
3	Revise and resubmit	ARCO APPROVED BY	DATE	SIGNATURE
4	Rejected	Anish Raj	03-Sep-2026	
5	For Information			

**DOCUMENT REVIEW STAMP**
(VENDOR/Subcontractor)



1. APPROVED / NO COMMENT

2. APPROVED WITH COMMENT – RESUBMIT

3. REVISE & RESUBMIT

4. REJECTED

5. INFORMATION

Acceptance in any of these categories in no way relieves the Vendor / Sub-Contractor of its responsibility for the due and proper performance of the works in accordance With the Purchase Order.

Review / Approval Code: 1

Reviewed / Approved by: Anish Raj 03-Sep-26



Vendor/Subcontractor to upload the native file under the Main Document section in next submission. Document section should be document itself (no crs), not Coversheet (Word file)

Abu Dhabi Company for Onshore Petroleum Operations Ltd.
(ADNOC ONSHORE)

EPC for SE AiP5 Project (On plot) - ASAB/SAHIL (Package 1)

AGREEMENT NO.: 4700025071
COMPANY PROJECT NO.: P30339B

INSPECTION AND TEST PLAN FOR CONTROL VALVES

DOCUMENT NO.: P30339B-30-99-52-4607

REVISION	DATE	PREPARED	REVIEWED	APPROVED	DESCRIPTION
2	02-SEP-26	STN	GNV	SH	Issued for Construction
1	11-AUG-26	STN	GNV	SH	Issued for Construction
B	19-JUN-26	STN	GNV	SH	Issued for Approval
A	06-NOV-25	STN	GNV	SH	Issued for Review

THIS DOCUMENT IS INTENDED FOR USE BY COMPANY AND ITS NOMINATED CONSULTANTS, CONTRACTORS, MANUFACTURERS AND SUPPLIERS

EPC CONTRACTOR  ARCHIRODON	SUBCONTRACTOR / VENDOR  KSB
ORIGINATOR DOCUMENT NO.: CD13E-ITP-CV-001	
ADNOC ONSHORE DOCUMENT NO. 30-99-52-4607	
Page 1 of 13	
ADNOC ONSHORE AGREEMENT NO. 4700025071	ADNOC ONSHORE PROJECT NO. P30339B
REV. 2	DATE: 02-SEP-26

INSPECTION AND TEST PLAN FOR CONTROL VALVES	Project No.: P30339B		
	ADNOC Onshore Doc. No.: 30-99-52-4607		
	Rev.: 2	Date: 02-SEP-26	
	Page 2 of 13		

SHEETS INDEX AND REVISIONS:

Rev. No.	Section	Revision Description
A	All	Issued for Review
B	All	Issued for Approval
1	All	Issued for Construction
2	All	Issued for Construction

HOLDS:

Hold No.	Section	Hold Description

INSPECTION AND TEST PLAN FOR CONTROL VALVES	Project No.: P30339B		
	ADNOC Onshore Doc. No.: 30-99-52-4607		
	Rev.: 2	Date: 02-SEP-26	
	Page 3 of 13		

Reference Documents

<u>SI No</u>	<u>Document Number</u>	<u>Document Name</u>
1	AGES-SP-04-002	Control Valve Specification
2	P30339B-14-01-40-4652	Datasheet for Control Valves – Asab (Batch 1)
3	P30339B-14-01-40-4658	Datasheet for Control Valves – Asab (Batch 2)
4	P30339B-16-01-40-4632	Datasheet for Control Valves – Sahil
5	P30339B-14-01-50-4643	General Arrangement Drawings for Control Valves – Asab (Batch 1)
6	P30339B-14-01-50-4649	General Arrangement Drawings for Control Valves – Asab (Batch 2)
7	P30339B-16-01-50-4635	General Arrangement Drawings for Control Valves – Sahil
8	P30339B-30-99-90-4699	Control Valves- Welding Procedures & Qualifications
9	P30339B-30-99-90-4701	Control Valves- NDE Procedure
10	P30339B-30-99-90-4703	Control Valves- Pressure Testing Procedures
11	P30339B-30-99-90-4705	Control Valves- PMI Procedure
12	P30339B-30-99-90-4707	Control Valves Painting/Coating Procedures (Valves + Actuator)
13	P30339B-30-99-90-4711	Fugitive Emission Test Procedures for Control Valves
14	P30339B-30-99-90-4712	Seat Leakage Test Procedure
15	P30339B-30-99-99-4602	Control Valves Technical Catalogues for Valves & All Accessories
16	P30339B-30-99-93-1679	Material Requisition for Control Valves
17	AGES-SP-07-004	Painting and Coating Specifications
18	P30339B-30-99-52-4608	Inspection and Test Plan for Self-Regulating Valves
19	P30339B-30-99-95-461	Control Valves factory Acceptance Test Procedure (FAT Procedure) (Valve + Actuator)

INSPECTION AND TEST PLAN FOR CONTROL VALVES	Project No.: P30339B		
	ADNOC Onshore Doc. No.: 30-99-52-4607		
	Rev.: 2	Date: 02-SEP-26	
	Page 4 of 13		

Testing Standards

<u>SI No</u>	<u>Standard Number</u>	<u>Standard Title</u>
1	ASME B 16.34	Design Standard
2	FCI 70.2	Testing Standard
3	ASME B 16.10	Face to face dimension standard
4	NACE MR0175-2021 ISO 15156-1 2020	NACE
5	ASME SEC V	NDE requirements
6	ASME SEC IX	Boiler and Pressure Vessel Code

	Project No:	P30339B	Project Name:	ADNOC SE AiP5 PROJECT (ON PLOT) – ASAB/SAHIL (PACKAGE 1)					
	Supplier	KSB MIL Controls Limited, Kerala, India KSB MIL Sale order number : CD13E (ITP CV-L2-4817 Rev 03)			Item / Equipment	Control Valves	Location:	INDIA	
	PO No.:	04108-PM-INST-008	Doc.No:	P30339B-30-99-52-4607	Rev.No.:	2	Page:	Page 5 of 13	

Inspection and Test Plan (ITP)																										
Criticality Rating	☒	1	☐	2	☐	3	☐	4	Certification	☒	3.2	☐	3.1	☐	2.2	☐	2.1	Inspection Level	☒	I	☐	II	☐	III	☐	IV

H: Hold; W10%: Witness on 10% per type,size ,rating; R: Review; M: Monitoring; A: Approval ; W : 100% Witness; RW-Random witness-10% witness on total order quantity per inspection offer lot.

Notification Period: Minimum of 14 days notice for “HOLD” and “WITNESS” points.

Act.No.:	Activity Description	Extent of Examination	Reference Document	Acceptance Criteria	Verifying Document	Quality Intervention					Remarks		
						Sub Vendor	KSB	TPIA (applicable where 3.2 certification required)	Archirodon	ADNOC Onshore			
1.1	Kick off Meeting	--	Purchase Order	Purchase Order	MOM	--	H	W	W	W	A	A	
1.2	Submission, Review & Approval of data sheet, GAD, ITP & all other Procedures as per VDRL	--	Customer PO, data sheets / GAD / ITP/ All documents as per VDRL	Customer PO, data sheets / GAD / ITP/ All documents as per VDRL	Customer PO, data sheets / GAD / ITP/ MOM	--	H	W	W	W	A	A	
1.3	Pre Inspection meeting	--	Datasheets, GAD, PO & Project Specifications	Datasheets, GAD, PO & Project Specifications	MOM	--	H	W	W	W	H		
2	INCOMING INSPECTION (BODY & BONNET)												
2.1	Body & Bonnet (Castings /Forgings)	Pouring of casting / Raw material identification for forgings	100%	Approved data sheet / ASTM A 216 Gr. WCC, ASTM A 352 Gr. LCC, ASTM A 494 CUSMCCu(N08826) & A 995 Gr.6A (CD3MWCuN) / Forging Equivalents AGES-SP-04-002, AGES-SP-09-003, AGES-SP-07-003/ NACE MR0175 / Approved GAD	Report	H	W	W	W	W	W	W	
2.2	Body & Bonnet (Castings /Forgings)	Receiving& Sampling of test bar	One sample per heat	Approved data sheet / ASTM A 216 Gr. WCC, ASTM A 352 Gr. LCC, ASTM A 494 CUSMCCu(N08826) & A 995 Gr.6A (CD3MWCuN) / Forging Equivalents AGES-SP-04-002, AGES-SP-09-003, AGES-SP-07-003/ NACE MR0175 / Approved GAD	Report	H	W	W	W	W	W	W	
2.3	Body & Bonnet (Castings /Forgings)	Heat treatment	100%	Approved data sheet / ASTM A 216 Gr. WCC, ASTM A 352 Gr. LCC, ASTM A 494 CUSMCCu(N08826) & A 995 Gr.6A (CD3MWCuN) / Forging Equivalents AGES-SP-04-002, AGES-SP-09-003, AGES-SP-07-003/ NACE MR0175 / Approved GAD	EN10204 3.2 Material Test certificate	H	R	R	R	R	R	R	
2.4	Body & Bonnet (Castings /Forgings)	Chemical Analysis	One sample per heat	Approved data sheet / ASTM A 216 Gr. WCC, ASTM A 352 Gr. LCC, ASTM A 494 CUSMCCu(N08826) & A 995 Gr.6A (CD3MWCuN) / Forging Equivalents AGES-SP-04-002, AGES-SP-09-003, AGES-SP-07-003/ NACE MR0175 / Approved GAD	EN10204 3.2 Material Test certificate	H	W	W	W	W	W	W	
2.5	Body & Bonnet (Castings /Forgings)	Mechanical Testing(Yield strength,Tensile stress, Elongation & Hardness check)	One sample per heat & heat treatment batch	Approved data sheet / ASTM A 216 Gr. WCC, ASTM A 352 Gr. LCC, ASTM A 494 CUSMCCu(N08826) & A 995 Gr.6A (CD3MWCuN) / Forging Equivalents AGES-SP-04-002, AGES-SP-09-003, AGES-SP-07-003/ NACE MR0175 / Approved GAD	EN10204 3.2 Material Test certificate	H	W	W	W	W	W	W	
2.6	Body & Bonnet (Castings /Forgings)	Impact Testing(at -46 deg celsius & As per datasheet)	One sample per heat & heat treatment batch	Approved data sheet / ASTM A 352 Gr. LCC / AGES-SP-04-002, AGES-SP-09-003, AGES-SP-07-003/ NACE MR0175 / Approved GAD	EN10204 3.2 Material Test certificate	H	W	W	W	W	W	W	
2.7	Body & Bonnet (Castings /Forgings)	Pitting Corrosion	One sample per heat & heat treatment batch	Approved data sheet / A 995 Gr.6A (CD3MWCuN) / A 494 CUSMCCu(N08826) / AGES-SP-04-002, AGES-SP-09-003, AGES-SP-07-003/ NACE MR0175 / Approved GAD For DSS material G48 method A shall be acceptance criteria(.For Inconel material G28 method A shall be acceptance criteria(Acceptance criteria of corrosion rate < 0.914 mm per year and no preferential attack in grain boundaries at 50 X magnification for inconel material)	EN10204 3.2 Material Test certificate	H	W	W	W	W	W	W	As per datasheet
2.8	Body & Bonnet (Castings /Forgings)	Surface Quality Check (For Castings)	100%	MSS SP 55 ASTM A 216 Gr. WCC, ASTM A 352 Gr. LCC, ASTM A 494 CUSMCCu(N08826) & A 995 Gr.6A (CD3MWCuN)	EN10204 3.2 Material Test certificate	H	W	W	W	W	W	W	

	Project No:	P30339B	Project Name:	ADNOC SE AiP5 PROJECT (ON PLOT) – ASAB/SAHIL (PACKAGE 1)					
	Supplier	KSB MIL Controls Limited, Kerala, India KSB MIL Sale order number : CD13E (ITP CV-L2-4817 Rev 03)			Item / Equipment	Control Valves	Location:	INDIA	
	PO No.:	04108-PM-INST-008	Doc.No:	P30339B-30-99-52-4607	Rev.No.:	2	Page:	Page 6 OF 13	

Inspection and Test Plan (ITP)																										
Criticality Rating	☒	1	☐	2	☐	3	☐	4	Certification	☒	3.2	☐	3.1	☐	2.2	☐	2.1	Inspection Level	☒	I	☐	II	☐	III	☐	IV

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Notification Period: Minimum of 14 days notice for “HOLD” and “WITNESS” points.

Act.No.:	Activity Description	Extent of Examination	Reference Document	Acceptance Criteria	Verifying Document	Quality Intervention					Remarks
						Sub Vendor	KSB	TPIA (applicable where 3.2 certification required)	Archirodon	ADNOC Onshore	
2.9	Body & Bonnet- ASTM A 216 Gr. WCC, ASTM A 352 Gr. LCC, ASTM A 494 CU5MCuC(N08826) & A 995 Gr.6A (CD3MWCuN) / Forging Equivalents (Castings /Forgings)	Non destructive testing for castings - Radiographic Testing at critical areas as per ASME B16.34 for castings. - Ultrasonic Testing for Forgings - DPT Test for Stainless Steel - MPT for Carbon steel /Alloy Steel (Wet Fluorescent)	As per approved data sheet	ASME Sec V, ASME B16.34/ Approved NDE Procedure/AGES-SP-07-003 & AGES-SP-09-003 AGES-SP-07-007	NDT Report & RT Film	H	W	W	W	W	
3	INCOMING INSPECTION (TRIM PARTS)										
3.1	Trims (Plug, Seat, Cage,Plug Stem as applicable)	Receiving& Sampling of test bar(applicable for castings and forgings)	100%	Approved data sheet / ASTM A 351 Gr. CF8M / ASTM A 494 CU5MCuC / ASTM B425 UNS N08825 / A 995 Gr.6A (CD3MWCuN) / SS316 / SS316L / ASTM B443 UNS N06625 / DUPLEX S32760 / Inconel 718 / INCONEL 825 / A 182 GR F61 / AGES-SP-04-002, AGES-SP-09-003, NACE MR0175 / AGES-SP-07-003/ Approved GAD	Report	H	W	W	W	W	During raw material identification stage,Heat number punching on the rod shall be correlated by TPI with respect to ROD Material TC and traceability shall be ensured.
3.2	Trims (Plug, Seat, Cage,Plug Stem as applicable)	Heat Treatment	100%	Approved data sheet / ASTM A 351 Gr. CF8M / ASTM A 494 CU5MCuC / ASTM B425 UNS N08825 / A 995 Gr.6A (CD3MWCuN) / SS316 / SS316L / ASTM B443 UNS N06625 / DUPLEX S32760 / Inconel 718 / INCONEL 825 / A 182 GR F61 / AGES-SP-04-002, AGES-SP-09-003, NACE MR0175 / AGES-SP-07-003/ Approved GAD	EN10204 3.2 Material Test certificate	H	R	R	R	R	3.2 certification is NA for rods.For rods/bar EN10204 3.1 MTC shall be issued and shall be tested at NABL accredited lab.3.1 TC For trim parts shall be from Bar stock (from original manufacturer or ISO /IEC/NABL approved LAB).

		Project No:	P30339B		Project Name:	ADNOC SE AiP5 PROJECT (ON PLOT) – ASAB/SAHIL (PACKAGE 1)						
		Supplier	KSB MIL Controls Limited, Kerala, India KSB MIL Sale order number : CD13E (ITP CV-L2-4817 Rev 03)				Item / Equipment	Control Valves	Location:	INDIA		
		PO No.:	04108-PM-INST-008		Doc.No:	P30339B-30-99-52-4607	Rev.No.:	2	Page:	Page 7 OF 13		

Inspection and Test Plan (ITP)																										
Criticality Rating	☒	1	☐	2	☐	3	☐	4	Certification	☒	3.2	☐	3.1	☐	2.2	☐	2.1	Inspection Level	☒	I	☐	II	☐	III	☐	IV

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Notification Period: Minimum of 14 days notice for “HOLD” and “WITNESS” points.

Act.No.:	Activity Description	Extent of Examination	Reference Document	Acceptance Criteria	Verifying Document	Quality Intervention					Remarks
						Sub Vendor	KSB	TPIA (applicable where 3.2 certification required)	Archirodon	ADNOC Onshore	
3.3	Trims (Plug, Seat, Cage,Plug Stem as applicable)	Chemical properties	One sample per heat	Approved data sheet / ASTM A 351 Gr. CF8M / ASTM A 494 CU5MCuC / ASTM B425 UNS N08825 / A 995 Gr.6A (CD3MWCuN) / SS316 / SS316L / ASTM B443 UNS N06625 / DUPLEX S32760 / Inconel 718 / INCONEL 825 / A 182 GR F61 / AGES-SP-04-002, AGES-SP-09-003, NACE MR0175 / AGES-SP-07-003/ Approved GAD	EN10204 3.2 Material Test certificate	H	W	W	W	W	3.2 certification is NA for rods.For rods/bar EN10204 3.1 MTC shall be issued and shall be tested at NABL accredited lab.3.1 TC For trim parts shall be from Bar stock (from original manufacturer or ISO /IEC/NABL approved LAB).
3.4	Trims (Plug, Seat, Cage,Plug Stem as applicable)	Mechanical Testing(Yield strength,Tensile stress, Elongation & Hardness check)	One sample per heat & heat treatment batch	Approved data sheet / ASTM A 351 Gr. CF8M / ASTM A 494 CU5MCuC / ASTM B425 UNS N08825 / A 995 Gr.6A (CD3MWCuN) / SS316 / SS316L / ASTM B443 UNS N06625 / DUPLEX S32760 / Inconel 718 / INCONEL 825 / A 182 GR F61 / AGES-SP-04-002, AGES-SP-09-003, NACE MR0175 / AGES-SP-07-003/ Approved GAD	EN10204 3.2 Material Test certificate	H	W	W	W	W	
3.5	Trims (Plug, Seat, Cage,Plug Stem as applicable)	Pitting Corrosion	One sample per heat & heat treatment batch	For DSS material G48 method A For Inconel material G28 method A Approved data sheet / A 995 Gr.6A (CD3MWCuN) / DUPLEX S32760 / AGES-SP-04-002, AGES-SP-09-003, NACE MR0175 / AGES-SP-07-003/ Approved GAD	EN10204 3.2 Material Test certificate	H	R	R	R	R	
3.6	Trims (Plug, Seat, Cage,Plug Stem as applicable)	Intergranular corrosion test as per ASTM A262 practice E for austenitic stainless materials	One sample per heat & heat treatment batch	Approved data sheet / ASTM A 351 Gr. CF8M / SS316 / SS316L / AGES-SP-04-002, AGES-SP-09-003, NACE MR0175 / AGES-SP-07-003/ Approved GAD	EN10204 3.2 Material Test certificate/Report	H	R	R	R	R	
4	INCOMING INSPECTION (CLADDING)										
4.1	Cladding(Body, Bonnet)	Material Conformity / Approved WPS	100%	Approved data sheet / Approved GAD / ASTM standards / MESC SPE 77/302 version 2020/ AGES-SP-07-007	Report	H	R	–	R	R	
4.2	Inspection and testing after cladding 1.LPT(Liquid Penetrant Testing) 2.UT(Ultrasonic Testing)- Disbondment - Partial UT shall be done wherever 100% UT is not feasible due to space constraints. 3.Visual&Thickness measurement 4.PMI test 5.Hardness		100%	Approved WPS / Approved data sheet / Approved GAD / ASTM standards / AGES-SP-07-007/AGES-SP-09-003 & AGES-SP-07-003	Reports	H	W	W	W	W	

		Project No:	P30339B		Project Name:	ADNOC SE AiP5 PROJECT (ON PLOT) – ASAB/SAHIL (PACKAGE 1)																				
		Supplier	KSB MIL Controls Limited, Kerala, India KSB MIL Sale order number : CD13E (ITP CV-L2-4817 Rev 03)				Item / Equipment	Control Valves	Location:	INDIA																
		PO No.:	04108-PM-INST-008		Doc.No:	P30339B-30-99-52-4607	Rev.No.:	2	Page:	Page 8OF 13																
Inspection and Test Plan (ITP)																										
Criticality Rating	☒	1	☐	2	☐	3	☐	4	Certification	☒	3.2	☐	3.1	☐	2.2	☐	2.1	Inspection Level	☒	I	☐	II	☐	III	☐	IV
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Notification Period: Minimum of 14 days notice for “HOLD” and “WITNESS” points.																										
Act.No.:	Activity Description	Extent of Examination	Reference Document	Acceptance Criteria	Verifying Document	Quality Intervention					Remarks															
						Sub Vendor	KSB	TPIA (applicable where 3.2 certification required)	Archirodon	ADNOC Onshore																
5	INCOMING INSPECTION (BOUGHT OUT ITEMS)																									
5.1	Body - Bonnet Fasteners	Material Conformity/ PO/ MR/ Project Specs	100%	Approved data sheet / ASTM standards / AGES-SP-09-005	Approved data sheet / ASTM standards	EN 10204 3.1 MTC required for fasteners	H	R	R	R	R															
5.2	Actuator spring	Material Conformity/ PO/ MR/ Project Specs	100%	Approved data sheet / ASTM standards	Approved data sheet / ASTM standards	EN 10204 3.1 MTC	H	R	R	R	R															
5.3	Accessories Smart positioner, Air Filter Regulator, Pressure Gauge, Relief Valve, Air filter, Solenoid Valve, Limit switch & Volume booster)	Hazardous area certificate,Degree of protection (if specified in datasheet)	Per Type	Approved data sheet and Approved GAD	Approved data sheet	TYP/IECEX/IP 66 Certification	H	R	R	R	R															
5.4	Certification for positioner	Material Conformity	Per Type	Approved data sheet/ Approved gad/ PO/ MR/ Project Specs/ International standards	Approved data sheet	Report	H	R	R	R	R															
6	IN PROCESS INSPECTION																									
6.1	Body, Bonnet & Trims after machining	a) Shell pressure Test	100%	ASME B16.34 / ISA S 75.19 (No Leak at 1.5 times rated working pressure) / AGES-SP-04-002, IEC-60534-4 / Approved GAD / Approved test procedure	ASME B16.34 / ISA S 75.19 (No Leak at 1.5 times rated working pressure)	TC	--	H	W	W	W															
6.2		b) DPT after machining	100%	ASME Sec V, ASME B16.34/ Approved NDE Procedure	ASME Sec V, ASME B16.34/ Approved NDE Procedure	Report	--	H	W (10%)	W (10%)	W (10%)															
6.3		c) Material check - PMI (Applicable for body, bonnet and trim parts as per datasheet. Test will be done with portable XRF)	100%	ASTM standards / Approved procedure	ASTM standards / Approved procedure	Report	--	H	W (10%)	W (10%)	W (10%)															
6.4		d) Hardness check on Hard-faced components (Stellite 6)	On Sample Pad	ASME Sec IX	WPS / WPQ / PQR / ASME Sec IX, Stellite thickness 1.9 mm minimum after final machining	Report	--	H	R	R	R															
7	ASSEMBLY TEST (VALVE WITH ACTUATOR)																									
7.1	Assembled Valve	a) Body mount Leakage test	100%	No Leak at 110% of rated pressure with water / Approved test procedure	No Leak at 110% of rated pressure with water / Approved test procedure	TC	--	H	W	W	W															
7.2		b) Seat Leakage	100%	FCI 70.2 / Leak class as per approved data sheet / Approved test procedure	FCI 70.2 / Leak class as per approved data sheet / Approved test procedure	TC	--	H	W	W	W															
7.3	Actuator Chamber	a) Strength & Leakage	100%	Leak check at 1.2 times of actuator supply pressure	No Leak at 1.2 times of actuator supply pressure	TC	--	H	W	W	W															
		b) Pneumatic circuit leak test(Tubing Leak)	100%	Leak check at 1.2 times of actuator supply pressure	No Leak at 1.2 times of actuator supply pressure	TC	--	H	W	W	W															

		Project No:	P30339B		Project Name:	ADNOC SE AiP5 PROJECT (ON PLOT) – ASAB/SAHIL (PACKAGE 1)																				
		Supplier	KSB MIL Controls Limited, Kerala, India KSB MIL Sale order number : CD13E (ITP CV-L2-4817 Rev 03)				Item / Equipment	Control Valves	Location:	INDIA																
		PO No.:	04108-PM-INST-008		Doc.No:	P30339B-30-99-52-4607	Rev.No.:	2	Page:	Page 9 OF 13																
Inspection and Test Plan (ITP)																										
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Act.No.:	Activity Description	Extent of Examination	Reference Document	Acceptance Criteria	Verifying Document	Quality Intervention					Remarks															
						Sub Vendor	KSB	TPIA (applicable where 3.2 certification required)	Archirodon	ADNOC Onshore																
7.4	Operational Test	c) Stroke Checking / Opening & Closing Time	100%	Approved data sheet	Approved data sheet	TC	--	H	W	W	W															
7.5		b) Linearity	100%	Approved test procedure / AGES-SP-04-002, IEC-60534-4	Approved test procedure / AGES-SP-04-002, IEC-60534-4	TC	--	H	W	W	W															
7.6		c) Hysteresis	100%	Approved test procedure / AGES-SP-04-002, IEC-60534-4	Approved test procedure / AGES-SP-04-002, IEC-60534-4	TC	--	H	W	W	W															
7.7		d) Accessories/Positioner calibration	100%	Approved data sheet /AGES-SP-04-002, IEC-60534-4	Approved data sheet	-	--	H	W	W	W	Positioner calibration shall be done along with valve performance.														
7.8		e) Failure action test / Dead band test	100%	Approved test procedure / AGES-SP-04-002, IEC-60534-4	Approved test procedure / AGES-SP-04-002, IEC-60534-4	TC	--	H	W	W	W															
7.9		e) Dead band check	100%	Approved test procedure / AGES-SP-04-002, IEC-60534-4	Approved test procedure / AGES-SP-04-002, IEC-60534-4	TC	--	H	W	W	W															
7.10		f) Functional test of position transmitters, limit switches, SOV as applicable.	100%	Approved data sheet / AGES-SP-04-002, IEC-60534-4	Approved data sheet / AGES-SP-04-002, IEC-60534-4	TC	--	H	W	W	W															
7.11	Fugitive Emission type test (Review of test certificate)	Leakage	One per type	Approved Procedure/ As per ISO 15848-1 / As per datasheet	Approved Procedure/ As per ISO 15848-1 / As per datasheet	FE type test report	--	H	R	A	A															
7.12	Fugitive Emission Production test	Leakage	One valve per type. sampling as per ISO 15848-2.	Approved Procedure/ As per ISO 15848-2 / As per datasheet	Approved Procedure/ As per ISO 15848-2 Limit Class B/ As per datasheet	TC	--	H	W	W	W															
8	Painting and allied activities																									
8.1	Surface preparation and Painting	Paint Material Selection	Each paint system	AGES-SP-07-004, ISO 12944-9	Specification and Product Data Sheet	Paint Manufacturer Compatibility Tests/Letter	--	H	R	R	R															
8.2	Surface preparation and Painting	Pre-production / Repair Qualification Test	PQT for each paint system	AGES-SP-07-004/ASTM D4228	Specification and Product Data Sheet Requirements/PQT shall be done on sample pad of dimension 300mm*300mm	RFI / Report	--	H	R	R	R															
8.3	Surface preparation and Painting	Incoming Abrasives / Paint Material inspection (Verify storage, container integrity, and shelf life)	Every Batch	ASTM D4940, Data sheet AGES-SP-07-004/ Visual	Specification and Product Data Sheet Requirments,Storage as per Paint datasheet(TDS) Conductivity ≤ 150 μmS/cm shall be measured at ambient temperature.	RFI / Report	--	H	R	R	R															
8.4	Surface preparation and Painting	Environmental Conditions	Before starting, Every 2 hours	ISO 8502-4	As per clause 12.1 of AGES-SP-07-004	RFI / Report	--	H	R	R	R															

	Project No:	P30339B		Project Name:	ADNOC SE AiP5 PROJECT (ON PLOT) – ASAB/SAHIL (PACKAGE 1)					
	Supplier	KSB MIL Controls Limited, Kerala, India KSB MIL Sale order number : CD13E (ITP CV-L2-4817 Rev 03)				Item / Equipment	Control Valves	Location:	INDIA	
	PO No.:	04108-PM-INST-008		Doc.No:	P30339B-30-99-52-4607	Rev.No.:	2	Page:	Page 10 OF 13	

Inspection and Test Plan (ITP)																										
Criticality Rating	☒	1	☐	2	☐	3	☐	4	Certification	☒	3.2	☐	3.1	☐	2.2	☐	2.1	Inspection Level	☒	I	☐	II	☐	III	☐	IV

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Act.No.:	Activity Description	Extent of Examination	Reference Document	Acceptance Criteria	Verifying Document	Quality Intervention					Remarks	
						Sub Vendor	KSB	TPIA (applicable where 3.2 certification required)	Archirodon	ADNOC Onshore		
8.5	Surface preparation and Painting	Pre-cleaning of surfaces / Solvent cleaning	100%	SSPC-SP 1	Free of oil, grease and other contaminants	RFI / Report	--	H	R	R	R	
8.6	Surface preparation and Painting	Compressed Air Quality	Every 2 hours / Twice per day	ASTM D4285	Free of oil and moisture	RFI / Report	--	H	R	R	R	
8.7	Surface preparation and Painting	Blasting /Surface preparation checking	100%	SSPC-VIS 1	Conform to the grade specified in the applicable Painting System	RFI / Report	--	H	RW	RW	RW	
8.8	Surface preparation and Painting	Roughness / Profile & Masking	Spot checks	NACE SP0287	Conform to the grade specified in the applicable painting system	RFI / Report	--	H	R	R	R	
8.9	Surface preparation and Painting	Dust test	Spot checks	ISO 8502-3	Maximum rating (1), Dust class (2)	RFI / Report	--	H	R	R	R	
8.10	Surface preparation and Painting	Water Soluble Salts	Spot checks	ISO 8502-6/9	≤ 2 µg/cm2 (20 mg/m2) BRESLE kit procedure only.	RFI / Report	--	H	RW	RW	R	
8.11	Surface preparation and Painting	Verify correct Mixing Ratio	100%	Manufacturer Data sheet	Mixing shall be completed using factory proportioned containers and as per paint manufacturer recommendation only	RFI / Report	--	H	R	R	R	
8.12	Surface preparation and Painting	WFT	Methodically during application	Metallic Comb Gauge, ISO 2808	According to specified DFT FOR EACH COAT	RFI / Report	--	H	R	R	R	
8.13	Surface preparation and Painting	Final DFT / Visual Examination	100% of surfaces	ISO 19840 / Visual and ISO 4628-1 to 6	Specified DFT FOR EACH COAT /No sagging, contaminations, orange peel, cracking, blistering, rust, damages and any other defects(Witness applicable for only final DFT,other Stages(primer,intermediate dft shall be reviewed only)	RFI / Report	--	H	W (10%) (Refer Note 5)	W (10%) (Refer Note 5)	W (10%) (Refer Note 5)	

	Project No:	P30339B		Project Name:	ADNOC SE AiP5 PROJECT (ON PLOT) – ASAB/SAHIL (PACKAGE 1)					
	Supplier	KSB MIL Controls Limited, Kerala, India KSB MIL Sale order number : CD13E (ITP CV-L2-4817 Rev 03)				Item / Equipment	Control Valves	Location:	INDIA	
	PO No.:	04108-PM-INST-008		Doc.No:	P30339B-30-99-52-4607	Rev.No.:	2	Page:	Page 11 OF 13	

Inspection and Test Plan (ITP)																										
Criticality Rating	☒	1	☐	2	☐	3	☐	4	Certification	☒	3.2	☐	3.1	☐	2.2	☐	2.1	Inspection Level	☒	I	☐	II	☐	III	☐	IV

H: Hold; W10%: Witness on 10% per type,size ,rating; R: Review; M: Monitoring; A: Approval ; W : 100% Witness; RW-Random witness-10% witness on total order quantity per inspection offer lot.

Notification Period: Minimum of 14 days notice for “HOLD” and “WITNESS” points.

Act.No.:	Activity Description	Extent of Examination	Reference Document	Acceptance Criteria	Verifying Document	Quality Intervention					Remarks	
						Sub Vendor	KSB	TPIA (applicable where 3.2 certification required)	Archirodon	ADNOC Onshore		
8.14	Surface preparation and Painting	(if applicable) Curing of Inorganic Zinc Silicate Primer using MEK solvent Rub Test	Spot checks	ASTM D4752	Conform to Level 5	RFI / Report	--	H	W (10%)	W (10%)	W (10%)	
8.15	Surface preparation and Painting	Holiday Testing	One sample per paint scheme	DFT ≤ 500 LV Wet sponge	Free of holidays and pinholes	RFI / Report	--	H	R	R	R	
8.16	Surface preparation and Painting	Adhesion Testing	Aleast 3 locations -one per paint scheme on sample pad	ISO 4624 (pull-of test) ASTM D4541	To be performed on test panels Minimum value > 5 MPa for Zinc primer-based and >7 MPa for non-Zinc based primer system	RFI / Report	--	H	W	W	W	
9	FINAL INSPECTION											
9.1	Overall dimension, Name plate details, Visual & Marking	Visual & Dimensions	100%	Approved data sheet & drawings	Approved data sheet & drawings	Report	--	H	W	W	W	
9.2	Packing	Soundness of Packing	100%	Approved Procedure	Approved Procedure	Report	--	H	W (10%)	W (10%)	W (10%)	
9.3	Product 3.2 certification	Review	100%	Approved data sheet & drawings	Approved data sheet & drawings	3.2 COC	--	H	R	R	R	
9.4	Inspection Release	Issue of IRN	100%	Customer's P.O./ Inspection documents	Customer's P.O./ Inspection documents	IRN	--	H	A	A	A	
9.5	Final MRB	MRB review & approval	100%	Valve test reports & Inspection documents.MRB Index	Valve test reports & Inspection documents	MRB	--	H	A	A	A	

All inspection and testing activities completed as per the requirements above, and accordingly the Inspection Release Note Issued.

Notes :

1) Certification shall be as per EN10204 Type 3.2 for castings / forgings. 3.2 certification is not applicable for components made from bars or rods. For such items, chemical,mechanical,hardness, corrosion tests (as applicable) will be retested at ISO/IEC accredited lab and shall be offered for witness.

2) Stock Material Request Guidelines doc no 30-99-97-1612 shall be followed.

3) For DSS supplier shall be NORSOK M650 or ISO 17782

4) Scope of Corrosion test witness (wherever applicable) shall be loading and unloading only.

5) Witness shall be 10% per type per size per rating and paint scheme.

Signed on Behalf of:	Sub-vendor	Vendor	TPIA	ARCHIRODON / Representative (TPIA)	ADNOC Onshore / Representative
Name:	--	Gautham Sai S			
Date:	--	11-08-2026			
Signature	--				

ANNEXURE-1

ITEM NO	TAG NO	VALVE SERIES	VALVE SIZE (IN)	BODY MATERIAL	BODY RATING	AREA
1	16-01-LCV-2803-01A	41300	3	ASTM A 494 CU5MCuC(N08826)	600	SAHIL
2	16-01-LCV-2803-01B	41300	3	ASTM A 494 CU5MCuC(N08826)	600	SAHIL
3	16-01-LCV-2803-36A	41100	2	ASTM A 494 CU5MCuC(N08826)	600	SAHIL
4	16-01-LCV-2803-36B	41100	2	ASTM A 494 CU5MCuC(N08826)	600	SAHIL
5	16-01-LCV-2803-38A	41100	2	ASTM A 494 CU5MCuC(N08826)	600	SAHIL
6	16-01-LCV-2803-38B	41100	2	ASTM A 494 CU5MCuC(N08826)	600	SAHIL
7	16-01-LCV-2812-01A	41300	3	ASTM A 494 CU5MCuC(N08826)	600	SAHIL
8	16-01-LCV-2812-01B	41300	3	ASTM A 494 CU5MCuC(N08826)	600	SAHIL
9	16-01-LCV-2813-01A	41100	2	ASTM A 494 CU5MCuC(N08826)	1500	SAHIL
10	16-01-LCV-2813-01B	41100	2	ASTM A 494 CU5MCuC(N08826)	1500	SAHIL
11	16-01-PCV-2405-03A	21100	2	ASTM A 216 Gr. WCC	300	SAHIL
12	16-01-PCV-2405-03B	21100	6	ASTM A 216 Gr. WCC	300	SAHIL
13	16-01-PCV-2803-37A	41300	8	ASTM A 352 Gr. LCC	600	SAHIL
14	16-01-PCV-2803-37B	41300	8	ASTM A 352 Gr. LCC	600	SAHIL
15	16-01-TCV-2811-02A	41900	14	ASTM A 352 Gr. LCC	300	SAHIL
16	16-01-TCV-2811-02B	41900	18	ASTM A 352 Gr. LCC	300	SAHIL
17	16-01-TCV-2812-18A	41900	12	ASTM A 352 Gr. LCC	600	SAHIL
18	16-01-TCV-2812-18B	41900	12	ASTM A 352 Gr. LCC	600	SAHIL
19	16-01-TCV-2813-58A	41900	8	ASTM A 352 Gr. LCC	1500	SAHIL
20	16-01-TCV-2813-58B	41900	10	ASTM A 352 Gr. LCC	1500	SAHIL
21	14-01-FCV-0200-03A	41600	14	ASTM A 216 Gr. WCC	600	ASAB
22	14-01-FCV-0200-03B	41600	14	ASTM A 216 Gr. WCC	600	ASAB
23	14-01-FCV-0200-04A	41600	16	ASTM A 216 Gr. WCC	900	ASAB
24	14-01-FCV-0200-04B	41600	16	ASTM A 216 Gr. WCC	900	ASAB
25	14-01-FCV-0305-07	33300	24	ASTM A 352 Gr. LCC + CLAD UNS N06625	300	ASAB
26	14-01-FCV-0305-09	33300	20	ASTM A 216 Gr. WCC	300	ASAB
27	14-01-FCV-0705-04A	21700	4	ASTM A 216 Gr. WCC	300	ASAB
28	14-01-FCV-0705-04B	41600	20	ASTM A 216 Gr. WCC	300	ASAB
29	14-01-FCV-0709-01	41600	10	ASTM A 216 Gr. WCC	300	ASAB
30	14-01-FCV-0710-01	41600	10	ASTM A 216 Gr. WCC	300	ASAB
31	14-01-FCV-0907-01	41600	8	ASTM A 216 Gr. WCC	300	ASAB
32	14-01-FCV-1006-01	91600	6	ASTM A 216 Gr. WCC	600	ASAB
33	14-01-FCV-1007-01	91600	6	ASTM A 216 Gr. WCC	600	ASAB
34	14-01-FCV-1225-01	41600	8	ASTM A 216 Gr. WCC	300	ASAB
35	14-01-FCV-1226-01	41600	8	ASTM A 216 Gr. WCC	300	ASAB
36	14-01-FCV-1300-02A	41600	12	ASTM A 216 Gr. WCC	900	ASAB
37	14-01-FCV-1300-02B	41600	12	ASTM A 216 Gr. WCC	900	ASAB
38	14-01-FCV-1300-03A	41600	12	ASTM A 216 Gr. WCC	900	ASAB
39	14-01-FCV-1300-03B	41600	12	ASTM A 216 Gr. WCC	900	ASAB
40	14-01-FCV-2435-01	21100	6	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	300	ASAB
41	14-01-FCV-2436-01	21100	6	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	300	ASAB
42	14-01-FCV-2437-01	78000	6	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	1500	ASAB
43	14-01-FCV-2438-01	78000	6	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	1500	ASAB
44	14-01-FCV-2439-01	21100	6	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	300	ASAB
45	14-01-FCV-2440-01	78000	6	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	1500	ASAB
46	14-01-LCV-0305-01	41600	12	ASTM A 216 Gr. WCC	300	ASAB
47	14-01-LCV-0305-02	41600	16	ASTM A 352 Gr. LCC + CLAD UNS N06625	300	ASAB
48	14-01-LCV-0405-02	21100	6	ASTM A 352 Gr. LCC + CLAD UNS N06625	300	ASAB
49	14-01-LCV-0709-01	41600	14	ASTM A 216 Gr. WCC	300	ASAB
50	14-01-PCV-0305-01	33300	30	ASTM A 216 Gr. WCC	300	ASAB
51	14-01-PCV-0305-02	41300	20	ASTM A 216 Gr. WCC	300	ASAB
52	14-01-PCV-0305-09	41300	20	ASTM A 216 Gr. WCC	300	ASAB
53	14-01-PCV-0405-01	41300	12	ASTM A 216 Gr. WCC	300	ASAB
54	14-01-PCV-0405-04	33300	14	ASTM A 216 Gr. WCC	300	ASAB

55	14-01-PCV-0705-07	33300	24	ASTM A 216 Gr. WCC	300	ASAB
56	14-01-PCV-0705-08	33300	30	ASTM A 216 Gr. WCC	300	ASAB
57	14-01-PCV -0804-08A	41300	6	ASTM A 216 Gr. WCC	300	ASAB
58	14-01-PCV-0804-08B	33300	12	ASTM A 216 Gr. WCC	300	ASAB
59	14-01-PCV-1105-01A	21700	4	ASTM A 216 Gr. WCC	300	ASAB
60	14-01-PCV-1105-01B	33300	10	ASTM A 216 Gr. WCC	300	ASAB
61	14-01-PCV-3700-01A	33300	30	ASTM A 216 Gr. WCC	300	ASAB
62	14-01-PCV-3700-01B	33300	30	ASTM A 216 Gr. WCC	300	ASAB
63	14-01-TCV-0410-45	41900	12	ASTM A 216 Gr. WCC	300	ASAB
64	14-01-HCV-0305-01	41600	2	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	300	ASAB
65	14-01-HCV-0305-02	21100	3	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	300	ASAB
66	14-01-HCV-0405-01	21100	3	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	300	ASAB
67	14-01-HCV -0804-01	21100	3	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	300	ASAB
68	14-01-HCV -1105-01	41600	2	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	300	ASAB
69	14-01-FCV-6006-01	41900	16	Duplex-SS (A 995 Gr.6A (CD3MWCuN))	300	ASAB
70	14-01-PCV-1004-17	41900	18	ASTM A 216 Gr. WCC	600	ASAB
71	14-01-FCV-1601-01A	41600	24	ASTM A 216 Gr. WCC	600	ASAB
72	14-01-FCV-1601-01B	41600	24	ASTM A 216 Gr. WCC	600	ASAB
73	14-01-FCV-0200-01A	41600	14	ASTM A 216 Gr. WCC	600	ASAB
74	14-01-FCV-0200-01B	41600	14	ASTM A 216 Gr. WCC	600	ASAB
75	16-01-PCV-3100-13	41600	3	ASTM A 352 Gr. LCC	1500	ASAB